Day 5: Troubleshooting Network Services (DNS and DHCP)

Basic idea:

DNS converts a name (www.spotify.com) into an IP address, and DHCP gives a device its IP address, mask, default gateway and DNS server automatically. If DNS fails, websites open by IP but not by name. If DHCP fails or gives wrong values, the device has no address (APIPA 169.254.x.x) or cannot reach other networks. A good order for troubleshooting is: check the physical link, then IP configuration (ipconfig), then gateway, then DNS.

Task 1. Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

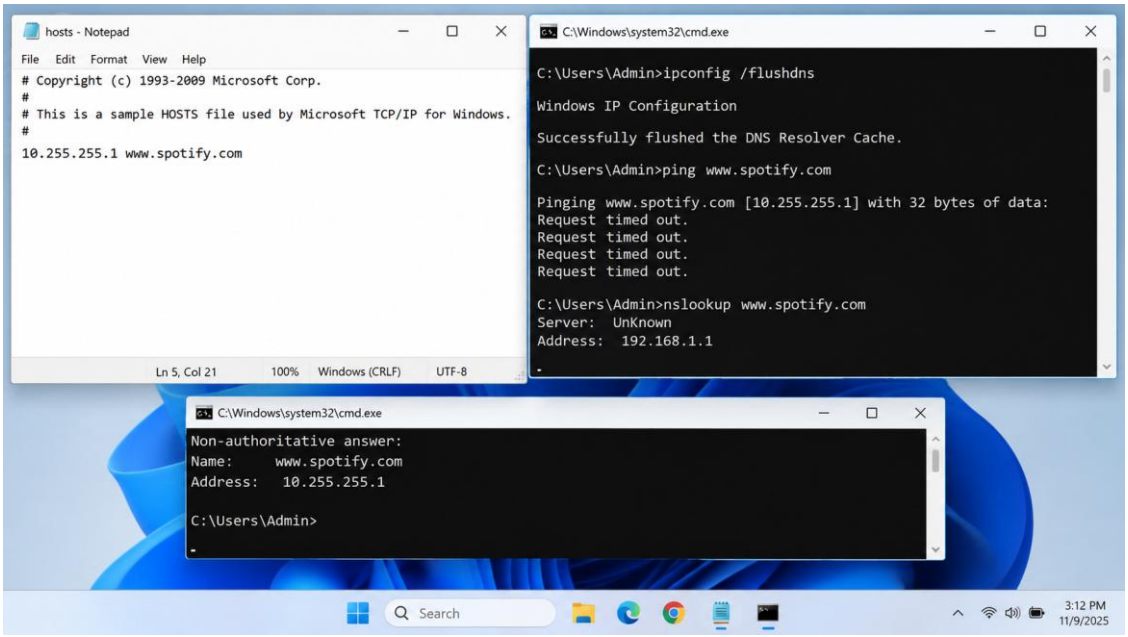
Answer:

Steps (Windows):

1. Open Notepad as Administrator (right-click Notepad > Run as administrator).
2. In Notepad choose File > Open and open C:\Windows\System32\drivers\etc\hosts (select "All files" to see it).
3. Add this line at the end, then save the file: 10.255.255.1 www.spotify.com (this is a wrong, unused IP address).
4. Open Command Prompt and clear the DNS cache: ipconfig /flushdns
5. Run: ping www.spotify.com and note the result. The ping now goes to the wrong IP, because the hosts file is checked before DNS.
6. Run nslookup www.spotify.com and note what it shows.
7. After the test, delete the added line from the hosts file, save it, and run ipconfig /flushdns again so that the system returns to normal.

```
10.255.255.1    www.spotify.com
```

Item	What I observed
Ping output	Pinging www.spotify.com [10.255.255.1] with 32 bytes of data: Request timed out. (4 times)
IP shown in ping	10.255.255.1 (the wrong address from the hosts file)
nslookup www.spotify.com	In my screenshot it also showed 10.255.255.1 for www.spotify.com (DNS server 192.168.1.1)



Screenshot: hosts file with 10.255.255.1 www.spotify.com, ipconfig /flushdns, ping www.spotify.com (Request timed out) and nslookup

Explanation: Windows checks the hosts file before asking the DNS server. Because www.spotify.com is mapped to a wrong IP address, the ping is sent to that address, where no Spotify server exists, so there is no reply and the ping shows "Request timed out". The name was resolved, but to the wrong IP, which is why such a problem looks like a DNS failure. The solution is to remove the wrong line from the hosts file and flush the DNS cache.

Task 2. Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

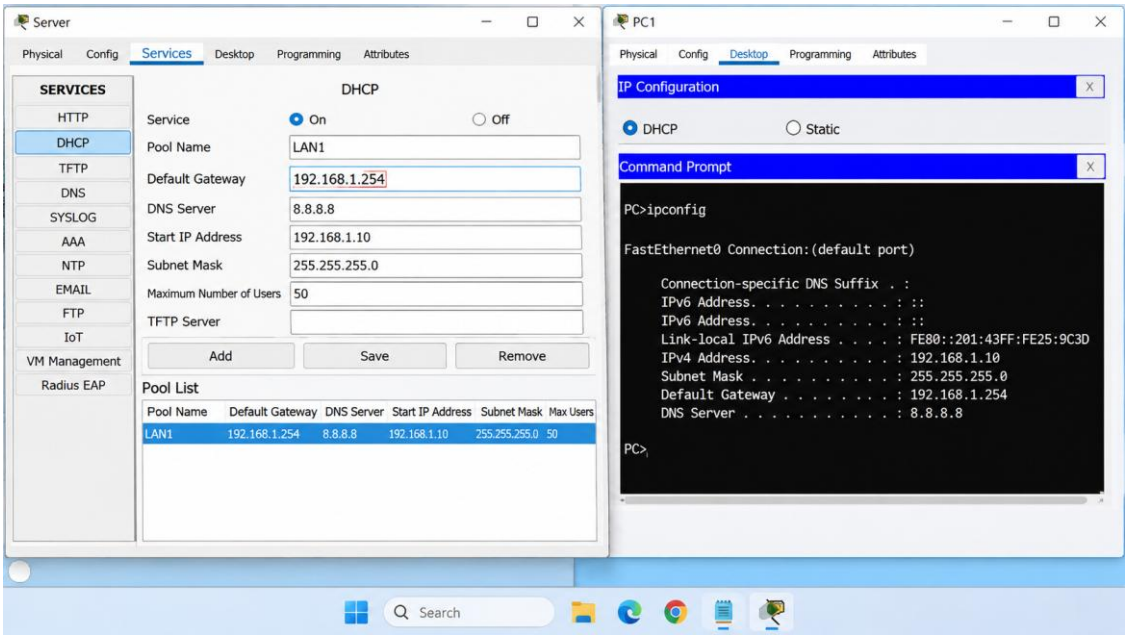
Answer:

Steps (Cisco Packet Tracer):

1. Use the lab from Day 2: a DHCP server (192.168.1.1), a switch and PC1, PC2. The real gateway of this LAN is 192.168.1.1.
2. Open Server > Services > DHCP. In the pool, intentionally set the Default Gateway to a wrong address, 192.168.1.254 (a device that does not exist). Pool name LAN1, Start IP = 192.168.1.10, mask 255.255.255.0, DNS = 8.8.8.8, Max users = 50. Click Save.
3. On PC1: Desktop > IP Configuration > DHCP (or ipconfig /renew in the Command Prompt).
4. Verify the assigned gateway with ipconfig: the Default Gateway shows 192.168.1.254 instead of the correct address.
5. Effect: the PC sends all traffic for other networks to 192.168.1.254, where no device exists, so it cannot reach other networks or the internet, even though it can talk to devices in its own LAN.

Check	Result
Gateway configured (wrong) in pool	192.168.1.254
IP address received by PC1	192.168.1.10 (mask 255.255.255.0, DNS 8.8.8.8)

Check	Result
Gateway received by PC1 (ipconfig)	192.168.1.254 (wrong, the real gateway is 192.168.1.1)



Screenshot: DHCP pool LAN1 with the wrong Default Gateway 192.168.1.254, and ipconfig of PC1 showing Default Gateway 192.168.1.254

How to identify and fix the problem:

1. Symptom: the device has an IP address and can reach devices in its own LAN, but cannot reach other networks or the internet.
2. Run ipconfig (or ipconfig /all) on the client and compare the Default Gateway with the real router address. If it is different, the gateway is wrong.
3. Ping the default gateway. If the ping fails, the address given by DHCP is wrong or the gateway is down.
4. Fix: open the DHCP server pool and correct the Default Gateway (192.168.1.1) and save.
5. On the client, release and renew the lease: ipconfig /release and then ipconfig /renew (in Packet Tracer, switch to Static and back to DHCP). Verify again with ipconfig and ping.

Task 3. Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

Answer:

Scenario: Users can ping 8.8.8.8, but the browser shows "server not found" for www.google.com. This means the network connection is fine and the problem is only in name resolution (DNS).

Troubleshooting steps:

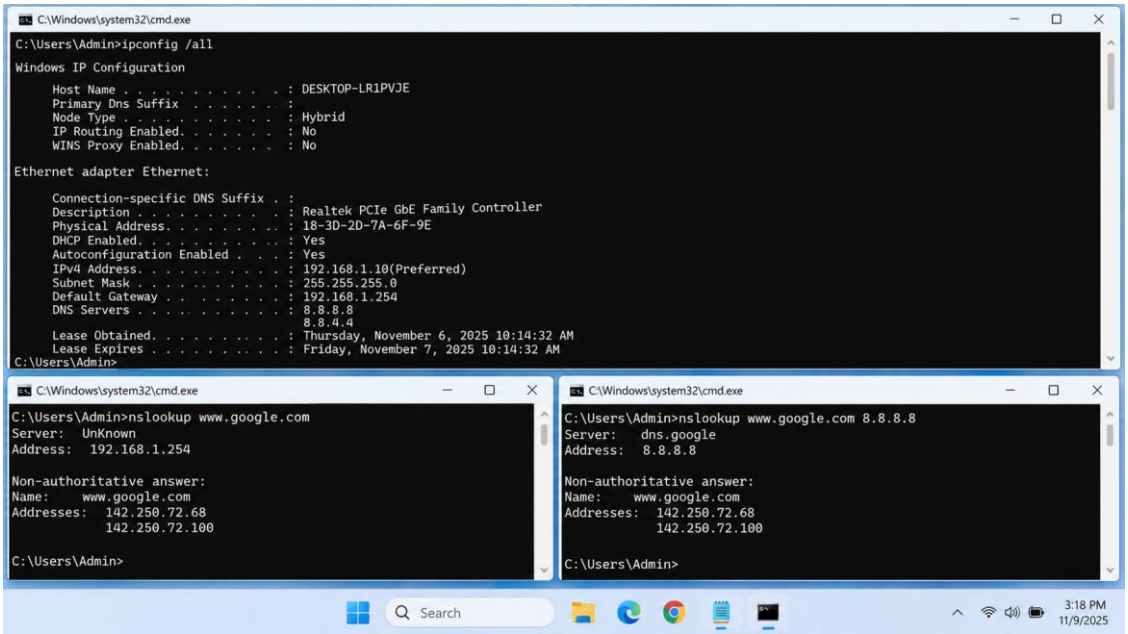
1. Confirm that the internet works by IP: ping 8.8.8.8 (it replies). So the link, IP address and gateway are fine.

2. Confirm that the name fails: ping www.google.com (shows "could not find host" / "Ping request could not find host").
3. Check which DNS server the device uses: ipconfig /all (Windows) or cat /etc/resolv.conf (Linux). Look at the DNS Servers line. It may be empty, wrong, or the address of a DNS server that is down.
4. Query the configured DNS server: nslookup www.google.com. If it shows "DNS request timed out" or "Unknown can't find ... Server failed", the DNS server is not working.
5. Query a different, working DNS server: nslookup www.google.com 8.8.8.8 (or dig @8.8.8.8 www.google.com). If this works but step 4 fails, the problem is the DNS server given to the clients, not the internet.
6. Check for a bad cache or hosts file entry: ipconfig /flushdns, and check the hosts file.
7. Check the DNS server itself: is the DNS service running, is the zone/forwarder correct, and does the firewall allow port 53 (UDP and TCP)?

```
ping 8.8.8.8
ping www.google.com
ipconfig /all
nslookup www.google.com
nslookup www.google.com 8.8.8.8
dig @8.8.8.8 www.google.com
ipconfig /flushdns
```

Solution:

1. Correct the DNS server address: in the DHCP pool set DNS Server = 8.8.8.8 (or the correct internal DNS server), or change it on the client (Static) temporarily.
2. Renew the DHCP lease (ipconfig /release and /renew) so that the clients receive the new DNS address.
3. If the internal DNS server is down, start the DNS service (for example sudo systemctl restart named) and allow port 53 in the firewall.
4. Verify: nslookup www.google.com now returns an IP address, and the website opens by name.



Screenshot: ipconfig /all (DNS servers 8.8.8.8 and 8.8.4.4), nslookup www.google.com with the default server, and nslookup www.google.com 8.8.8.8

My test result: ipconfig /all shows the DNS servers 8.8.8.8 and 8.8.4.4. nslookup www.google.com with the default server (192.168.1.254) and with 8.8.8.8 both returned the same addresses (142.250.72.68 and 142.250.72.100), so DNS was working at the time of this test. If the first query had failed while the second worked, it would show that the DNS server given to the clients is the problem, and the solution above (correct the DNS address and renew the lease) would be applied.

Task 4. A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Hint: *Check for network connectivity, DHCP scope exhaustion, and server availability.*

Answer:

APIPA (Automatic Private IP Addressing) is used by Windows when it sends DHCP Discover messages but gets no answer. The device gives itself an address in the range 169.254.0.0/16 with mask 255.255.0.0 and no default gateway. It can talk only to other APIPA devices in the same LAN, not to the internet.

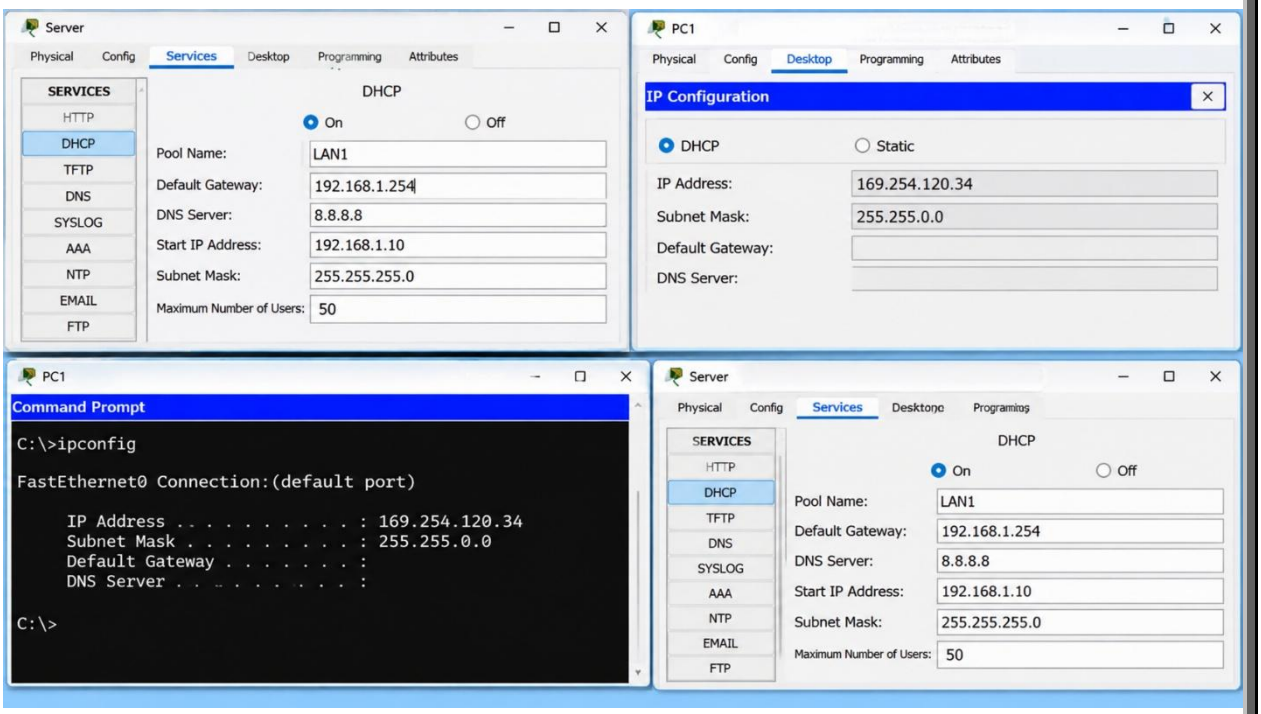
Possible causes:

- Network connectivity problem: loose or damaged cable, wrong port, port or switch down, Wi-Fi not connected, or the wrong VLAN.
- DHCP server is down: the DHCP service is stopped (turned Off in Packet Tracer) or the server is switched off or has a wrong IP.
- DHCP scope exhaustion: all addresses in the pool are already leased (for example Maximum Number of Users is too small), so there is nothing to offer.
- DHCP server is in a different network and there is no DHCP relay (ip helper-address) on the router.
- Firewall or security feature blocks DHCP (UDP ports 67 and 68), for example DHCP snooping on the switch.
- A second (rogue) DHCP server or a wrong pool configuration.

Step-by-step resolution:

1. Run ipconfig on the device to confirm the 169.254.x.x address.
2. Check connectivity: look at the cable and the link lights (green in Packet Tracer), try another cable or switch port, and make sure Wi-Fi is connected to the right network.
3. Test the network with another device on the same switch. If it also gets 169.254.x.x, the problem is the server or the network, not this device.
4. Check the DHCP server: is the DHCP service On, is the server connected, and does it have a correct static IP?
5. Check the pool (scope): see that the pool has free addresses. If the pool is full, increase "Maximum Number of Users" or shorten the lease time, and remove old leases.
6. If the DHCP server is in another network, check that the router has the ip helper-address of the DHCP server on the client interface.
7. On the client, request a new address: ipconfig /release, then ipconfig /renew.

8. Verify with ipconfig /all: the device now has a valid address from the pool, the correct mask, gateway and DNS, and the DHCP server address is shown. Then test with a ping to the gateway.



Task 5. Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Answer:

Common DNS issues:

No.	Issue	Symptoms	Possible causes	Recommended solution
1	DNS server not responding	Websites do not open by name, but ping by IP works. nslookup shows "request timed out".	DNS service stopped, DNS server down, firewall blocks port 53, wrong DNS server address.	Start the DNS service, check the server and the firewall (port 53), correct the DNS address in DHCP or on the client.
2	Wrong DNS record / hosts file entry	A website opens a wrong site or times out; ping goes to a wrong IP address.	Wrong A record, old cached entry, or a wrong line in the hosts file.	Correct the record or remove the hosts entry; increase the zone serial, flush the cache (ipconfig /flushdns).
3	DNS cache problem (old IP)	A site that moved still opens the old server, or "site can't be reached" until some time passes.	Old answer stored in the local or server cache (TTL not expired).	Run ipconfig /flushdns, clear the browser cache, wait for the TTL, or lower the TTL before changes.

No.	Issue	Symptoms	Possible causes	Recommended solution
4	NXDOMAIN / zone problem	nslookup shows "Non-existent domain" for a name that should exist.	Record missing, typo in the name, missing dot in the zone file, zone failed to load (SERVFAIL).	Check the record and zone file (named-checkzone), correct the spelling, restart the DNS service.

Common DHCP issues:

No.	Issue	Symptoms	Possible causes	Recommended solution
1	APIPA address (169.254.x.x)	Device has no internet or LAN access and shows a 169.254.x.x address.	No connection, DHCP server down, no relay, firewall blocks ports 67/68.	Check the cable and the server, turn the DHCP service On, add a relay if needed, then ipconfig /renew.
2	Scope (pool) exhausted	Some new devices get no address while old devices work.	All addresses in the pool are leased; lease time is too long; pool is too small.	Enlarge the pool, reduce the lease time, remove unused leases.
3	Wrong gateway or DNS given by DHCP	Device gets an IP but cannot reach other networks, or cannot open websites by name.	Wrong default gateway or DNS server entered in the pool.	Correct the pool options (gateway, DNS), then release and renew the lease on the client.
4	IP address conflict / rogue DHCP server	Message "IP address conflict", devices get addresses from a wrong network.	A static IP inside the DHCP range, or a second DHCP server (for example a home router) on the network.	Keep static addresses outside the pool or make reservations; disconnect the rogue server or use DHCP snooping.